

Avinash Tiwari, Ph.D. Student

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Current Affiliations

- 2021 – 📌 Inter-University Centre for Astronomy and Astrophysics, Pune.
- 2022 – 📌 LIGO Scientific Collaboration and LIGO-India Scientific Collaboration (LISC).

Education

- 2021 – 📌 **Ph.D.**, Inter-University Centre for Astronomy and Astrophysics, Pune.
- 2019 – 2021 📌 **M.Sc. Physics**, Indian Institute of Technology Guwahati, Guwahati.
- 2016 – 2019 📌 **B.Sc. Physics, Mathematics, and Computer Science**, University of Allahabad, Prayagraj.

Research Background

- 📌 **Data Analysis:** Bayesian parameter estimation, Fisher matrix analysis, and hierarchical inference.
- 📌 **Waveform Modeling & Environments:** Analytical modeling of line-of-sight acceleration and velocity waveforms, and probing CBC environments.
- 📌 **Cosmology:** Early universe cosmology driven by the synergies between gravitational-wave observations and 21-cm signals.

Skills

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|--------------------------------|---|
| Languages | 📌 English, Hindi. |
| Programming & Analytical Tools | 📌 Python, C, $\LaTeX$, Mathematica. |
| Bayesian Inference & Sampling | 📌 BILBY, BILBY_PIPE, BILBY_TGR, GWPOPULATION, EMCEE, DYNESTY. |
| Computing & Infrastructure | 📌 High-Performance Computing, HTCondor. |
| Open Source Contributions | 📌 Core developer for the Line-of-Sight Acceleration (LOSA) pipeline within the BILBY_TGR package. |

Awards & Honors

- 📌 **Junior Research Fellowship (JRF):** Awarded by the Council of Scientific and Industrial Research (CSIR), Government of India (All India Rank 89 in Physical Sciences). (2020).
- 📌 **Institute Merit-cum-Means Scholarship:** Awarded by the Indian Institute of Technology Guwahati for academic excellence (top 25% of the M.Sc. cohort). (2019 – 2021).

Conferences, Workshops, and Colloquia

- 📌 **GW - Detector Characterization Workshop** (December 2025), IUCAA, Pune.
- 📌 **GR24 - Amaldi16** (July 2025). Talk: “Profiling stellar environments of gravitational wave sources”.
- 📌 **AEI Colloquium** (August 2024), AEI Hanover: “Profiling stellar environments of gravitational wave sources”.

Conferences, Workshops, and Colloquia (continued)

- **New ideas on the ORIGIN of BLACK HOLE MERGERS** (August 2024), NBI Copenhagen. **Invited Talk:** “Profiling stellar environments of gravitational wave sources.”
- **ICGC-2023** (December 2023). Talk: “Probing globular clusters and other astrophysical environments with gravitational waves emitted by accelerated compact binary mergers.”
- **Amaldi15** (July 2023). Talk: “Probing globular clusters and other astrophysical environments with gravitational waves emitted by accelerated compact binary mergers”.
- **Discussion Meeting on LUNAR GRAVITATIONAL-WAVE DETECTION** (April 17 - 20, 2023), Ramanujan Lecture Hall, ICTS Bengaluru.
- **IAGRG32** (December 2022). Poster presentation and a Flash Talk: “Waltzing binaries: probing line-of-sight acceleration of merging compact objects with gravitational waves”.

Other Talks / Seminars

- **Last Friday Talks at IUCAA** (August 29, 2025): “Profiling Dark Matter Spikes with Gravitational Waves from Accelerated Binaries”.
- **Last Friday Talks at IUCAA** (September 27, 2024): “Profiling stellar environments of gravitational wave sources”.
- **Last Friday Talks at IUCAA** (July 28, 2023): “Probing globular clusters and other astrophysical environments with gravitational waves emitted by accelerated compact binary mergers”.
- **Last Friday Talks at IUCAA** (March 31, 2023): “Waltzing binaries: probing line-of-sight acceleration of merging compact objects with gravitational waves”.
- **Online Seminar at ICTS, Bengaluru** (December 15, 2022): “Waltzing binaries: probing line-of-sight acceleration of merging compact objects with gravitational waves”.

Courses Tutored

- PHY-IC481 - General Relativity at IUCAA.
- PHY-IE485 - Elective course - Cosmology at IUCAA.

Mentoring

- Sudhir Gholap
- Md Asim Ansari

Journal Referee

- Monthly Notices of the Royal Astronomical Society

References

- **Prof. Shasvath J. Kapadia**, IUCAA, shasvath.j.kapadia@gmail.com
- **Dr. Aditya Vijaykumar**, CITA, aditya@utoronto.ca
- **Prof. Sourav Chatterjee**, TIFR, souravchatterjee.tifr@gmail.com
- **Prof. Basudeb Dasgupta**, TIFR, bdasgupta@theory.tifr.res.in
- **Prof. Susmita Adhikari**, IISRE Pune, susmita@iiserpune.ac.in
- **Prof. Tirthankar Roy Choudhury**, NCRA-TIFR, troych75@gmail.com